

PostgreSQL ↔ SQL Server

End-to-End DBA Correlation Guide

A practical PostgreSQL learning and administration reference for experienced SQL Server DBAs

Important: These mappings are conceptual. PostgreSQL and SQL Server often use different implementation architectures even when the DBA-facing purpose is similar. Where there is no true one-to-one equivalent, this guide explicitly says so.

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1. Core Architecture

#	PostgreSQL	SQL Server	Easy DBA Understanding
1	PostgreSQL Server	SQL Server Database Engine	Main database engine
2	Cluster	SQL Server Instance	Collection of databases managed by one PostgreSQL server/data directory; not a SQL Server Failover Cluster
3	Database	Database	Similar logical container
4	Schema	Schema	Very similar
5	Table	Table	Same basic concept
6	View	View	Same
7	Materialized View	Indexed View	Similar purpose; implementation differs
8	Function	Function / Stored Procedure	PostgreSQL functions are heavily used
9	Procedure	Stored Procedure	Similar
10	Extension	SQL Server component/feature	Adds functionality
11	WAL	Transaction Log	Critical mapping
12	shared_buffers	Buffer Pool	Closest conceptual mapping
13	work_mem	Query workspace memory	Similar purpose, not identical
14	maintenance_work_mem	Maintenance memory	VACUUM/index operations
15	Background writer	Background/checkpoint processes	Writes dirty buffers
16	Checkpointer	Checkpoint	Similar responsibility
17	WAL writer	Log Writer	Writes WAL records
18	Autovacuum	No direct equivalent	PostgreSQL-specific critical subsystem
19	VACUUM	No direct equivalent	Storage/concurrency maintenance
20	MVCC	Row-versioning/MVCC	Implementation differs
21	Visibility Map	No direct equivalent	PostgreSQL storage optimization
22	Free Space Map	No direct equivalent	Tracks reusable space

2. Installation & Services

PostgreSQL	SQL Server	DBA Mapping
PostgreSQL package	SQL Server installation media/package	Software installation
postgres OS user	SQL Server service account	Service identity
postgresql.service	SQL Server service	Start/stop engine
systemctl start/stop/status	Start/Stop/Get-Service	Service control
pg_isclusters	Configuration Manager / instance discovery	Cluster discovery
pg_ctl	SQL Server control utilities	Server control
psql	SSMS/sqlcmd	CLI client
pgAdmin	SSMS	GUI administration
5432	1433	Default TCP port
Multiple PostgreSQL clusters	Multiple SQL Server instances	Similar operational concept; different implementation

3. PostgreSQL Cluster vs SQL Server Instance

PostgreSQL	SQL Server	Comparison
Cluster	Instance	Closest conceptual mapping
Contains multiple databases	Contains multiple databases	Similar
Own data directory/port	Own instance configuration	Similar isolation
Not a failover cluster	FCI is an HA technology	Critical distinction

4. Configuration

PostgreSQL	SQL Server	Purpose
postgresql.conf	Server configuration	Main configuration
pg_hba.conf	Authentication/access configuration	Client authentication
listen_addresses	TCP/IP configuration	Listening interfaces
port=5432	TCP 1433	Network port
max_connections	user connections / connection limits	Concurrent connections
shared_buffers	Buffer Pool	Data-page cache
work_mem	Query memory grant concepts	Per-operation query memory
maintenance_work_mem	Maintenance memory	Maintenance operations
effective_cache_size	No direct equivalent	Planner cache assumption
random_page_cost	No direct equivalent	Planner cost model
wal_level	Recovery/replication configuration	WAL capability
max_wal_size	Log/checkpoint behavior	WAL sizing
checkpoint_timeout	Checkpoint configuration	Checkpoint frequency
autovacuum	No direct equivalent	Automatic vacuum/analyze
log_min_duration_statement	Query-duration logging / Extended Events	Slow query detection
log_statement	Audit / Extended Events	Statement logging

5. Authentication & Security

PostgreSQL	SQL Server
Role	Login/User depending on context
Login role	SQL Server Login
Database role	Database Role
pg_hba.conf	Authentication/access configuration
SCRAM-SHA-256	SQL authentication/password authentication
LDAP	AD/LDAP integration
Kerberos	Kerberos/Windows Authentication
SSL/TLS	TLS encryption
GRANT	GRANT
REVOKE	REVOKE
pg_roles	sys.server_principals
ALTER ROLE	ALTER LOGIN / ALTER USER
Role membership	Role membership
Row-Level Security	Row-Level Security
SECURITY DEFINER	EXECUTE AS concepts
search_path	Default schema/object resolution concepts

6. Database & Schema

PostgreSQL	SQL Server
Database	Database
Schema	Schema
public schema	dbo is closest common schema
CREATE DATABASE	CREATE DATABASE
CREATE SCHEMA	CREATE SCHEMA
Database owner	Database owner
Schema owner	Schema owner
Tablespaces	Filegroups
LC_COLLATE	Collation/locale concepts
CREATE EXTENSION	Install/enable feature/component

7. Storage Architecture

PostgreSQL	SQL Server
Data directory	SQL Server data directory
Relation	Database object/storage structure
Heap table	Heap
8-KB page	8-KB page
Tuple	Row
TOAST	No direct equivalent
FSM	No direct equivalent
Visibility Map	No direct equivalent
WAL	Transaction log
Tablespace	Filegroup
pg_relation_size()	sys.dm_db_partition_stats / sp_spaceused concepts
pg_total_relation_size()	Object/database space usage
pg_database_size()	Database size

8. Tables

PostgreSQL	SQL Server
CREATE TABLE	CREATE TABLE
Heap table	Heap
Primary key	Primary key
Foreign key	Foreign key
Unique constraint	Unique constraint
Check constraint	Check constraint
Default	Default constraint
Generated column	Computed column
Identity	IDENTITY
Serial	Legacy sequence-based auto-numbering
GENERATED ALWAYS AS IDENTITY	IDENTITY
Temporary table	#temp
Unlogged table	No direct equivalent
Partitioned table	Partitioned table
RETURNING	OUTPUT

9. Data Types

PostgreSQL	SQL Server	Notes
smallint	smallint	Same
integer	int	Same purpose
bigint	bigint	Same
numeric	decimal/numeric	Same family
real	real	Same name
double precision	float	Closest
money	money	Semantics differ
boolean	bit	Closest
char	char	Same
varchar	varchar	Same
text	varchar(max)	Conceptual
date	date	Same
time	time	Same
timestamp	datetime2	Closest
timestamp with time zone	datetimeoffset	Closest

uuid	uniqueidentifier	Closest
json	JSON in character column	Implementation differs
jsonb	No exact equivalent	PostgreSQL-native binary JSON
bytea	varbinary	Closest
array	No direct equivalent	Native PostgreSQL array
range types	No direct equivalent	PostgreSQL-specific
inet	No direct equivalent	Network address type

10. SQL Language

PostgreSQL	SQL Server	Easy Mapping
LIMIT	TOP	Row limiting
OFFSET ... LIMIT	OFFSET ... FETCH	Pagination
RETURNING	OUTPUT	Return affected rows
COALESCE	COALESCE	Same
NULLIF	NULLIF	Same
CASE	CASE	Same
STRING_AGG	STRING_AGG	Same
GENERATE_SERIES()	Recursive CTE/tally table	Series generation
FILTER	Conditional aggregation	Aggregate filtering
LATERAL	CROSS/OUTER APPLY	Closest concept
WITH	CTE	Same
Window functions	Window functions	Same
EXPLAIN	Estimated execution plan	Plan inspection
EXPLAIN ANALYZE	Actual execution plan	Execute and inspect

11. Stored Procedures & Functions

PostgreSQL	SQL Server
Function	Function
Procedure	Stored procedure
PL/pgSQL	T-SQL
CREATE FUNCTION	CREATE FUNCTION
CREATE PROCEDURE	CREATE PROCEDURE
CALL	EXEC
RAISE NOTICE	PRINT/logging
RAISE EXCEPTION	THROW
EXCEPTION block	TRY...CATCH
SECURITY DEFINER	EXECUTE AS
IMMUTABLE/STABLE/VOLATILE	No direct equivalents

12. Indexes

PostgreSQL	SQL Server	Mapping
B-tree	B-tree	Core index
Hash	Hash index concepts	Not equivalent in all scenarios
GiST	No direct equivalent	Specialized index framework
SP-GiST	No direct equivalent	Specialized
GIN	No direct equivalent	Common for JSONB/search use cases
BRIN	No direct equivalent	Large physically ordered tables
Partial index	Filtered index	Closest
Expression index	Indexed expression concept	Closest
Unique index	Unique index	Same
Multicolumn	Composite	Same purpose

INCLUDE	Included columns	Same purpose
REINDEX	ALTER INDEX REBUILD	Closest maintenance
pg_stat_user_indexes	sys.dm_db_index_usage_stats	Index usage

13. Partitioning

PostgreSQL	SQL Server	Mapping
Declarative partitioning	Table partitioning	Same broad feature
Range partition	RANGE partition	Same
List partition	LIST partition	Same
Hash partition	HASH partition	Same broad concept
Partition key	Partitioning column	Same
Partition pruning	Partition elimination	Same optimizer goal
Default partition	No exact equivalent	PostgreSQL feature
Attach partition	Add/switch-like operation	Conceptual
Detach partition	Remove/switch-like operation	Conceptual
pg_partition_tree()	Partition metadata DMVs	Metadata

14. Transactions

PostgreSQL	SQL Server
BEGIN	BEGIN TRANSACTION
COMMIT	COMMIT
ROLLBACK	ROLLBACK
SAVEPOINT	SAVE TRANSACTION
MVCC	Row-versioning/MVCC
Transaction ID/XID	Transaction metadata
Snapshot	Snapshot/isolation concepts
pg_stat_activity	sys.dm_exec_sessions / requests

15. Isolation Levels

PostgreSQL	SQL Server	Important
Read Uncommitted	READ UNCOMMITTED	Names map; behavior must be validated
Read Committed	READ COMMITTED	Implementation differs
Repeatable Read	REPEATABLE READ	Semantics differ
Serializable	SERIALIZABLE	Similar isolation goal
Snapshot-style behavior	SNAPSHOT	Implementation differs
PostgreSQL Repeatable Read	SQL Server Repeatable Read	Do not assume identical behavior

16. Locking & Blocking

PostgreSQL	SQL Server
pg_locks	sys.dm_tran_locks
pg_stat_activity	sys.dm_exec_requests/sessions
pg_blocking_pids()	Blocking-session DMV queries
pg_terminate_backend()	KILL
Row locks	Row/key locks
Table locks	Table locks
Advisory locks	sp_getapplock concept
Deadlock detection	Deadlock detection
PostgreSQL logs	Extended Events/error log

17. Query Performance

PostgreSQL	SQL Server
pg_stat_activity	sys.dm_exec_requests
pg_stat_statements	Query Store/DMVs
EXPLAIN	Execution plan
EXPLAIN ANALYZE	Actual execution plan
Planner	Query Optimizer
Sequential Scan	Table Scan
Index Scan	Index Seek/Scan depending on plan
Bitmap Heap Scan	No exact equivalent
Bitmap Index Scan	Bitmap operations in plans
Nested Loop	Nested Loops
Hash Join	Hash Match
Merge Join	Merge Join
Sort	Sort
Aggregate	Stream/Hash Aggregate

18. Execution Plan Translation

PostgreSQL	SQL Server
Seq Scan	Table Scan
Index Scan	Index Scan
Index Only Scan	Covering/index-only concept
Bitmap Heap Scan	No direct operator equivalent
Nested Loop	Nested Loops
Hash Join	Hash Match
Merge Join	Merge Join
Sort	Sort
Aggregate	Aggregate
Materialize	Spool/materialization concepts
Gather	Parallelism
Gather Merge	Parallelism + ordered merge
Parallel Seq Scan	Parallel Table Scan
Parallel Index Scan	Parallel Index Scan
Planning Time	Compile/optimization time
Execution Time	Query execution duration

19. Statistics

PostgreSQL	SQL Server
ANALYZE	UPDATE STATISTICS
Autovacuum Analyze	Auto statistics maintenance
pg_stats	sys.stats / statistics DMVs
Extended statistics	Multi-column statistics concepts
Histogram	Histogram
Most common values	Frequency information
default_statistics_target	Statistics sampling/configuration concepts
Planner statistics	Optimizer statistics

20. Vacuum & Bloat

PostgreSQL	SQL Server / DBA Translation
VACUUM	No direct equivalent Removes/reclaims dead tuples for reuse
VACUUM FULL	No exact equivalent Compacts/rebuilds table storage; disruptive

Autovacuum	No direct equivalent Automatic vacuum/analyze
Table bloat	Fragmentation/unused space, not identical Do not equate directly
Dead tuples	Obsolete row versions conceptually Different mechanism
Freeze	No equivalent Prevents XID wraparound
XID wraparound	No equivalent PostgreSQL-specific risk
pg_repack	Rebuild/online-maintenance concepts External tool

21. Backup

PostgreSQL	SQL Server
pg_dump	Logical backup/scripting
pg_dumpall	Instance-level logical scripting
pg_basebackup	Physical/base backup
Filesystem backup	File/database backup
WAL archive	Transaction log backup/archive stream
pg_restore	Restore logical backup
psql restore	Script execution

22. PITR

PostgreSQL	SQL Server
Base backup	Full backup
WAL archive	Transaction log backups
Recovery target time	STOPAT
Recovery target LSN	LSN-based recovery concepts
restore_command	Backup restore chain
recovery_target_time	STOPAT
recovery_target_lsn	LSN-oriented recovery
Timeline	No direct equivalent

23. WAL vs SQL Server Transaction Log

PostgreSQL WAL	SQL Server Transaction Log
WAL records	Log records
WAL segments	Transaction log structures
WAL archive	Log backup/archive
LSN	LSN
WAL sender	Replication sender concepts
WAL receiver	Secondary/log receiver concepts
pg_wal	Transaction log files
wal_level	Recovery/replication configuration
archive_mode	Log archive strategy
archive_command	Backup/archive mechanism
Replication slot	No exact equivalent

24. High Availability

PostgreSQL	SQL Server
Streaming Replication	AG/log-based secondary concepts
Physical Standby	AG/DR secondary
Synchronous replication	Synchronous commit
Asynchronous replication	Asynchronous commit
Hot Standby	Readable secondary
Primary	Primary replica

Standby	Secondary replica
Patroni	No direct equivalent
EFM	No direct equivalent
HA manager	AG/WSFC orchestration concepts

25. PostgreSQL HA Architecture

PostgreSQL Pattern	SQL Server Pattern
Application → HAProxy → Primary/Standby → Patroni → etcd/Consul	Application → AG Listener → Primary/Secondaries → WSFC/quorum
External load balancer commonly used	AG Listener is integrated
HA manager commonly needed for automatic failover	AG has integrated failover management

26. Replication

PostgreSQL	SQL Server
Physical replication	AG/log-shipping-like physical concepts
Logical replication	Transactional replication/CDC-like use cases
Publication	Publication
Subscription	Subscriber/subscription
Replication slot	No exact equivalent
WAL sender	Replication sender
WAL receiver	Replication receiver
Logical decoding	Change-data extraction concepts
Replication origin	No direct equivalent

27. SQL Server Always On AG vs PostgreSQL Streaming Replication

Area	PostgreSQL	SQL Server
Primary	Primary	Primary replica
Secondary	Standby	Secondary replica
Sync	Synchronous replication	Synchronous commit
Async	Asynchronous replication	Asynchronous commit
Read-only	Hot Standby	Readable secondary
Automatic failover	Usually HA manager required	AG functionality
Listener	External HA/load balancing commonly used	AG Listener
Quorum	DCS/HA-manager dependent	WSFC/quorum
Unit	Cluster/physical WAL stream	Database replica

28. Log Shipping

PostgreSQL	SQL Server
WAL archiving + restore	Log shipping
Archive WAL	Copy transaction log
Restore WAL	Restore transaction log
Standby recovery	Secondary database
restore_command	Log-shipping restore job
Recovery target	Standby recovery

29. Disaster Recovery

PostgreSQL	SQL Server
WAL archiving	Transaction log backups

PITR	Point-in-time restore
Streaming replica	AG secondary
Delayed standby	Delayed log shipping/DR concept
Cross-region replica	Cross-region AG/DR secondary
Object-storage WAL archive	Cloud backup storage
Base backup	Full backup
Timeline	No direct equivalent

30. Jobs & Scheduling

PostgreSQL	SQL Server
pgAgent	SQL Server Agent
cron	SQL Agent schedule / OS scheduler
systemd timer	SQL Agent schedule
pg_cron	SQL Server Agent
pgBackRest scheduled jobs	SQL Agent backup jobs
Shell scripts	PowerShell/Cmd
PL/pgSQL maintenance procedure	T-SQL maintenance procedure

31. Monitoring

PostgreSQL	SQL Server
pg_stat_activity	sys.dm_exec_requests/sessions
pg_stat_database	Database DMVs
pg_stat_user_tables	Table/index DMVs
pg_stat_user_indexes	sys.dm_db_index_usage_stats
pg_locks	sys.dm_tran_locks
pg_stat_bgwriter	Checkpoint/I/O diagnostics
pg_stat_wal	Transaction log/WAL diagnostics
pg_stat_replication	AG/replication DMVs
pg_replication_slots	Replication metadata
pg_stat_progress_vacuum	No direct equivalent
pg_stat_progress_create_index	Index operation monitoring
pg_stat_statements	Query Store
PostgreSQL logs	SQL Server error log
Prometheus/Grafana	SQL monitoring tools

32. SQL Server DBA Monitoring Translation

SQL Server	PostgreSQL
sp_who2	pg_stat_activity
sp_WhoIsActive	pg_stat_activity + custom queries
sys.dm_exec_requests	pg_stat_activity
sys.dm_exec_sessions	pg_stat_activity
sys.dm_tran_locks	pg_locks
Query Store	pg_stat_statements

33. Performance Tuning

SQL Server DBA Concept	PostgreSQL
CPU	CPU
Memory pressure	Memory/cache pressure
Buffer cache	shared_buffers + OS cache
Query Store	pg_stat_statements

Execution plan	EXPLAIN
Actual plan	EXPLAIN ANALYZE
Missing index DMVs	No direct equivalent
Statistics	PostgreSQL statistics
Parameter sniffing	Prepared-plan/generic-vs-custom behavior differs
Blocking	Blocking
Deadlocks	Deadlocks
Wait statistics	No direct equivalent
Query hints	Planner controls differ
Plan guides	No direct equivalent
Forced plans	Different tooling

34. Wait Statistics

SQL Server	PostgreSQL
sys.dm_os_wait_stats	No direct equivalent
PAGEIOLATCH/WRITELOG/CXPACKET/etc.	No one-to-one mapping
Wait stats analysis	wait_event_type + wait_event + pg_stat_activity + pg_stat_statements + OS metrics Different architecture

35. SQL Server TempDB vs PostgreSQL Temporary Storage

SQL Server	PostgreSQL
TempDB	Temporary relations/files
#temp	CREATE TEMP TABLE
##temp	No direct equivalent
Table variables	No direct equivalent
TempDB version store	PostgreSQL MVCC/storage architecture
TempDB contention	Different PostgreSQL temporary I/O considerations

36. Encryption

PostgreSQL	SQL Server
TLS/SSL	TLS
Storage/filesystem/cloud encryption	TDE
pgcrypto	Cryptographic functions
Column/application encryption	Always Encrypted / encryption functions
KMS integration	Key hierarchy/external key management
Certificates	Certificates

37. Auditing

PostgreSQL	SQL Server
pgaudit	SQL Server Audit
log_statement	Audit/Extended Events
log_connections	Login auditing
log_disconnections	Session auditing
Native logs	SQL Server logs
External SIEM	External SIEM

38. Database Mail / Notifications

SQL Server	PostgreSQL
Database Mail	No native exact equivalent

SQL Agent alerts	External monitoring
sp_send_dbmail	External SMTP/application/script
Operator notifications	Monitoring platform

39. SSIS Comparison

SQL Server	PostgreSQL Ecosystem
SSIS	External ETL tools
Data Flow Task	ETL pipeline
Control Flow	Workflow/orchestration
SSISDB	No direct equivalent
Package	ETL workflow
Script Task	Python/Bash/custom code
Execute SQL Task	SQL execution step
SQL Agent	Scheduler

40. SSRS Comparison

SQL Server	PostgreSQL
SSRS	External BI/reporting platform
Report Server	BI server
Report	Dashboard/report
Dataset	SQL query/data source
Paginated Report	Reporting tool
Subscription	BI/report scheduler

41. SQL Agent vs PostgreSQL Automation

SQL Server	PostgreSQL
SQL Server Agent	pg_cron / pgAgent / cron / systemd / external automation
Jobs	Scheduled tasks
Alerts	Monitoring/notification tooling
Maintenance plans	Scripts + scheduler

42. DBCC Commands

SQL Server	PostgreSQL
DBCC CHECKDB	No exact equivalent
DBCC CHECKTABLE	No direct equivalent
DBCC CHECKALLOC	No direct equivalent
DBCC SHRINKDATABASE	No direct equivalent
DBCC SHRINKFILE	No direct equivalent
DBCC FREEPROCCACHE	No direct equivalent
DBCC DROPCLEANBUFFERS	No direct equivalent

43. DBCC CHECKDB Equivalent

SQL Server	PostgreSQL
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DBCC CHECKDB	amcheck + catalog/storage validation + backup/restore validation
Index consistency	amcheck
Space/maintenance health	VACUUM/ANALYZE and storage checks

44. Database Maintenance

SQL Server	PostgreSQL
Index rebuild	REINDEX
Index reorganize	No exact equivalent
Update statistics	ANALYZE
DBCC CHECKDB	amcheck + other checks
Backup	pg_basebackup/pg_dump
Shrink database	Generally not routine
Cleanup old data	DELETE/partition management
Transaction log maintenance	WAL/archive management
Maintenance plan	pg_cron/pgAgent/scripts

45. Connection Pooling

PostgreSQL	SQL Server
PgBouncer	Application/middleware connection pooling
Session pooling	Connection pooling
Transaction pooling	No direct equivalent
Statement pooling	No direct equivalent
Backend process per connection	SQL Server worker/session architecture

46. Migration

PostgreSQL	SQL Server / Tools
pg_dump/pg_restore	Logical migration tools
Logical replication	Replication-based migration
AWS DMS	AWS DMS
Azure migration tooling	Azure migration tooling
ora2pg	Oracle migration tooling
postgres_fdw	Linked Server-like integration

47. SQL Server Linked Server vs FDW

PostgreSQL	SQL Server
postgres_fdw	Linked Server
tds_fdw/other FDWs	Linked Server/provider
Pushdown	Provider/optimizer dependent
Remote PostgreSQL	Linked Server

48. Database Links

Oracle	PostgreSQL	SQL Server
DB Link	dblink / FDW	Linked Server

49. Replication Slots

PostgreSQL	SQL Server
Replication slot	No direct equivalent
Purpose	Prevents WAL recycling while consumer needs it
Inactive slot	Can retain WAL
Operational risk	pg_wal growth and disk exhaustion

50. LSN

PostgreSQL	SQL Server
WAL LSN	Transaction log LSN
pg_current_wal_lsn()	Current log position concept
pg_wal_lsn_diff()	LSN difference concept
Replication lag	AG/log-shipping lag metrics

51. Replication Lag

PostgreSQL	SQL Server
Primary WAL position	Primary log position
Standby receive LSN	Secondary received
Standby replay LSN	Secondary redone/replayed
Operational question	How far behind is the secondary?

52. High Availability Managers

PostgreSQL	SQL Server
Patroni	AG/WSFC orchestration concepts
EFM	HA manager
repmgr	Replication/cluster management
pg_auto_failover	Automated failover management
etcd/Consul	Distributed consensus/configuration
HAProxy	Listener/load balancing role

53. Patroni vs SQL Server AG

Patroni/PostgreSQL	SQL Server AG
PostgreSQL replication	AG replication
Patroni manages failover	AG manages failover
DCS	WSFC/quorum architecture
PostgreSQL primary	AG primary
PostgreSQL replica	AG secondary
HAProxy	AG Listener/load balancer

54. Cloud Mapping

PostgreSQL	SQL Server
Azure Database for PostgreSQL	Azure SQL Database
Azure PostgreSQL Flexible Server	Azure SQL Managed Instance
PostgreSQL on Azure VM	SQL Server on Azure VM

Amazon RDS for PostgreSQL	Amazon RDS for SQL Server
Aurora PostgreSQL	No exact SQL Server equivalent
PostgreSQL on EC2	SQL Server on EC2
RDS Multi-AZ	SQL Server HA architecture
Read Replica	SQL read/replica concepts depending on platform

55. System Databases

PostgreSQL	SQL Server
postgres	No exact equivalent
template0	Model-like template concept
template1	Model-like template concept
pg_catalog	System catalog
information_schema	INFORMATION_SCHEMA
pg_toast	No equivalent
master/model/msdb/tempdb	SQL Server system databases Do not map one-to-one

56. Metadata

PostgreSQL	SQL Server
pg_catalog	sys catalog
information_schema	INFORMATION_SCHEMA
pg_class	Object metadata
pg_namespace	Schemas
pg_database	sys.databases
pg_roles	sys.server_principals
pg_tables	sys.tables
pg_indexes	sys.indexes
pg_attribute	sys.columns
pg_constraint	Constraint/object metadata
pg_stat_activity	DMVs

57. Common DBA Commands

Task	PostgreSQL	SQL Server
List databases	\l	sp_helpdb / sys.databases
Connect	\c dbname	USE dbname
List tables	\dt	sp_tables / sys.tables
Describe table	\d table	sp_help table
List users	\du	sys.server_principals
Current database	current_database()	DB_NAME()
Current user	current_user	SUSER_SNAME()
Server version	version()	@@VERSION
Kill session	pg_terminate_backend(pid)	KILL session_id
Activity	pg_stat_activity	sys.dm_exec_requests
Locks	pg_locks	sys.dm_tran_locks

58. PostgreSQL psql vs SQLCMD

PostgreSQL psql	SQL Server sqlcmd
\l	Database listing

\c db	USE db
\dt	Table listing
\d table	sp_help
\du	Login/role listing
\dn	Schema listing
\df	Function listing
\q	Exit
\timing	Query timing
\x	Expanded output

59. Error Handling

PostgreSQL	SQL Server
EXCEPTION block	TRY...CATCH
RAISE EXCEPTION	THROW
RAISE NOTICE	PRINT/logging
SQLSTATE	Error number/state concepts
GET STACKED DIAGNOSTICS	ERROR_*() functions

60. Dynamic SQL

PostgreSQL	SQL Server
EXECUTE in PL/pgSQL	EXEC / sp_executesql
format()	QUOTENAME/string construction
EXECUTE ... USING	sp_executesql parameters

61. Temporary Objects

PostgreSQL	SQL Server
CREATE TEMP TABLE	CREATE TABLE #temp
Temporary table	Local temp table
ON COMMIT DROP	No direct equivalent
Temporary schema	TempDB concept
Temporary files	TempDB-like temporary work; architecture differs

62. Identity / Sequence

PostgreSQL	SQL Server
SEQUENCE	SEQUENCE
nextval()	NEXT VALUE FOR
currval()	No direct exact equivalent
Identity	IDENTITY
Serial	Legacy sequence-based mechanism
Identity generated always	IDENTITY

63. Materialized Views

PostgreSQL	SQL Server
Materialized View	Indexed View
REFRESH MATERIALIZED VIEW	Automatic indexed-view maintenance
Manual refresh	Transactionally maintained
Can become stale	Indexed view generally maintained with base changes

64. JSON

PostgreSQL	SQL Server
json	JSON stored in text
jsonb	No exact equivalent
jsonb_path_query	JSON path functions
->	JSON_VALUE / JSON_QUERY concepts
->>	JSON_VALUE
GIN JSONB index	No direct equivalent
JSON containment	JSON querying

65. Full Text Search

PostgreSQL	SQL Server
tsvector	Full-Text Search
tsquery	Full-Text query
GIN/GIST	Full-text indexes
to_tsvector()	Full-text indexing concept
to_tsquery()	CONTAINS/FREETEXT concepts

66. Regular Expressions

PostgreSQL	SQL Server
~	No direct equivalent
~*	No direct equivalent
regexp_replace()	Complex T-SQL/custom approaches
regexp_matches()	No direct equivalent
Native regex	Historically limited native regex capability

67. Extensions

PostgreSQL	SQL Server
CREATE EXTENSION	Enable/install feature/component
pg_stat_statements	Query performance extension
pgcrypto	Cryptography extension
pgaudit	Auditing extension
PostGIS	Spatial extension
pg_trgm	Trigram search
uuid-oss	UUID functions
Built-in features/CLR	SQL Server extensibility concepts

68. Foreign Data Wrapper

PostgreSQL	SQL Server
postgres_fdw	Linked Server
Remote PostgreSQL	Linked Server
Remote SQL Server via FDW	Linked Server
Remote Oracle via FDW	Linked Server
Pushdown	Provider-dependent linked query optimization

69. Connection / Session Terminology

PostgreSQL	SQL Server
Backend PID	SPID/session
Session	Session
Connection	Connection
pg_stat_activity.pid	session_id
client_addr	Client/network information
application_name	Application name/program name
state	Request/session state

70. PostgreSQL Process Architecture

PostgreSQL	SQL Server
Process-oriented backend architecture	Worker/scheduler-oriented engine
Backend process per client session	Worker/session architecture
Checkpointer	Checkpoint processes
WAL writer	Log writer
Background writer	Background/checkpoint activity

71. Memory Architecture

PostgreSQL	SQL Server
shared_buffers	Buffer Pool
work_mem	Query workspace/memory grant concept
maintenance_work_mem	Maintenance memory
temp_buffers	Temporary relation buffers
effective_cache_size	Optimizer cache assumption
OS page cache	OS cache
Per-operation memory	Memory grant concepts

72. CPU & Parallelism

PostgreSQL	SQL Server
max_worker_processes	Worker/thread architecture
max_parallel_workers	Parallel worker concepts
max_parallel_workers_per_gather	Parallelism per query
min_parallel_table_scan_size	Cost-based parallelism
parallel_setup_cost	Parallelism cost model
parallel_tuple_cost	Parallelism cost model
MAXDOP	No direct single parameter
Cost Threshold	No direct equivalent

73. SQL Server Concepts Without Direct PostgreSQL Equivalents

SQL Server	PostgreSQL
SQL Server Agent	pg_cron/pgAgent/external scheduler
DBCC CHECKDB	No single equivalent
TempDB	No direct equivalent
AG Listener	External HA/load-balancing layer commonly used
Query Store	pg_stat_statements + tooling

Extended Events	Logging/extensions/observability
Database Mail	External notification system
SSIS	External ETL
SSRS	External reporting
Resource Governor	No direct equivalent
Plan Guides	No direct equivalent
TDE	No exact native equivalent
Filegroups	Tablespaces are conceptually closest
Differential backup	No direct native equivalent
Transaction log backup	WAL archiving
Indexed view	Materialized view; behavior differs
Filtered index	Partial index
FCI	No direct PostgreSQL equivalent

74. PostgreSQL Concepts SQL Server DBAs Must Learn Separately

PostgreSQL Concept	Why Important
MVCC internals	Fundamental concurrency model
VACUUM	Critical database health operation
Autovacuum	Critical production subsystem
XID	Transaction ID management
XID wraparound	Major operational risk
Freeze	Prevents transaction ID wraparound
WAL	Foundation of recovery/replication
Replication slots	Can cause WAL accumulation
Timeline	Important for recovery/replication
pg_hba.conf	Authentication architecture
search_path	Object resolution/security
TOAST	Large-value storage
GIN	Powerful index type
GiST	Specialized index framework
BRIN	Large-table optimization
Logical decoding	Change-data extraction
FDW	Remote data integration
Extensions	Major extensibility model
Patroni	Common HA management
PgBouncer	Connection pooling
pg_stat_statements	Core performance analysis

75. DBA Daily Operations Mapping

Daily Task	PostgreSQL	SQL Server
Check server	systemctl/PostgreSQL tools	SQL Server service
Connections	pg_stat_activity	DMVs
Blocking	pg_locks/pg_blocking_pids()	DMVs
Long queries	pg_stat_activity	DMVs/Query Store
Query history	pg_stat_statements	Query Store
Database size	pg_database_size()	sys.database_files/DMVs
Table size	pg_total_relation_size()	sp_spaceused
Indexes	pg_indexes	sys.indexes
Index usage	pg_stat_user_indexes	sys.dm_db_index_usage_stats
Statistics	pg_stats	sys.stats
Update stats	ANALYZE	UPDATE STATISTICS

Dead tuples	VACUUM	No direct equivalent
Rebuild index	REINDEX	ALTER INDEX REBUILD
Backup	pg_basebackup/pg_dump	BACKUP DATABASE
Restore	pg_restore/recovery	RESTORE DATABASE
PITR	WAL + base backup	Full + log backups
HA status	pg_stat_replication/Patroni	AG DMVs
Kill session	pg_terminate_backend()	KILL
Schedule job	pg_cron/pgAgent	SQL Agent

76. DBA Troubleshooting Translation

Problem	PostgreSQL First Check	SQL Server First Check
High CPU	pg_stat_activity, pg_stat_statements, OS CPU	DMVs/Query Store
Blocking	pg_locks, pg_stat_activity	DMVs
Deadlock	PostgreSQL logs	Extended Events/error log
Slow query	EXPLAIN ANALYZE, pg_stat_statements	Actual plan/Query Store
Table bloat	pg_stat_user_tables, VACUUM stats	Fragmentation/space
High WAL	pg_stat_wal, replication slots	Transaction log
Replica lag	pg_stat_replication	AG DMVs
Disk full	WAL/data/temp	Data/log/tempdb
Connection exhaustion	pg_stat_activity/max_connections/PgBouncer	Connection DMVs
Corruption	amcheck/logs/storage	DBCC CHECKDB
Statistics problem	ANALYZE/statistics	UPDATE STATISTICS
Index problem	pg_stat_user_indexes/EXPLAIN	DMVs/plan
Authentication	pg_hba.conf/logs	Logins/permissions/error log
Recovery failure	WAL/archive/recovery config	Backup chain/log restore

77. Backup Strategy Translation

PostgreSQL	SQL Server
BASE BACKUP + WAL ARCHIVE + RECOVERY	FULL + DIFF + LOG + PITR
Base backup	Full backup
WAL archive	Transaction log backups
Recovery target	STOPAT
No native Full→Differential→Log hierarchy	Native backup chain

78. HA/DR Strategy Translation

PostgreSQL	SQL Server
Application → HAProxy/LB → Primary/Standby → Patroni → DCS	Application → AG Listener → Primary/Secondaries → WSFC/quorum
External HA/load balancing common	Integrated AG Listener
Streaming WAL	Transaction log/AG data movement
Patroni failover	AG failover

79. Most Important Mental Translation Map

SQL Server DBA	PostgreSQL DBA
Instance	Cluster
Database	Database

Schema	Schema
Login	Role
User	Role/user
SQL Agent	pg_cron/pgAgent
SSMS	pgAdmin
sqlcmd	psql
T-SQL	SQL + PL/pgSQL
Transaction Log	WAL
Log Backup	WAL archiving
Full Backup	Base backup
PITR	PITR
Always On AG	Streaming Replication + HA manager
AG Secondary	Standby
AG Listener	HAProxy/load balancer/DNS/service endpoint
WSFC	HA manager + DCS concepts
Query Store	pg_stat_statements + tooling
DMVs	pg_stat_* / pg_catalog
Buffer Pool	shared_buffers + OS cache
TempDB	Temporary relations/files
DBCC CHECKDB	amcheck + validation
UPDATE STATISTICS	ANALYZE
Index Rebuild	REINDEX
Fragmentation	Bloat/dead tuples — not identical
Filtered Index	Partial Index
Indexed View	Materialized View — not identical
Linked Server	FDW
sp_who2	pg_stat_activity
KILL	pg_terminate_backend()
SQL Audit	pgaudit
TDE	Storage/filesystem/cloud encryption
SSIS	External ETL
SSRS	External reporting
Database Mail	External SMTP/monitoring
Filegroup	Tablespace — not identical
Partition Elimination	Partition Pruning
MAXDOP	PostgreSQL parallel-worker settings
CXPACKET/CXCONSUMER	No direct equivalent
Wait Statistics	wait_event + activity/statistics
Extended Events	Logging/observability
Replication Slot	No direct equivalent
Autovacuum	No SQL Server equivalent
XID Freeze	No SQL Server equivalent
TOAST	No SQL Server equivalent

80. Recommended Learning Sequence for a SQL Server DBA

80. Recommended Learning Sequence for a SQL Server DBA

Phase	PostgreSQL Topics	SQL Server Anchor
1	Architecture	SQL Server Engine architecture
2	Installation / services	SQL Server installation
3	postgresql.conf	Server configuration
4	pg_hba.conf	SQL authentication/security

5	Roles/privileges	Logins/users/roles
6	Database/schema/storage	Database/files/filegroups
7	Tables/data types	Tables/data types
8	Indexes	B-tree/indexes
9	EXPLAIN	Execution plans
10	Statistics/ANALYZE	Statistics
11	MVCC	SQL Server row-versioning
12	Locks/blocking	Lock manager
13	VACUUM/autovacuum	Maintenance/fragmentation
14	WAL	Transaction log
15	pg_dump	Logical backup
16	pg_basebackup	Physical/base backup
17	WAL archiving	Transaction log backups
18	PITR	SQL Server STOPAT
19	Streaming replication	Always On secondary
20	Replication slots	New PostgreSQL concept
21	Patroni	AG failover/orchestration
22	PgBouncer	Connection management
23	pg_stat_statements	Query Store
24	Performance tuning	SQL Server performance tuning
25	Partitioning	SQL Server partitioning
26	Logical replication	SQL Server replication
27	FDW	Linked Server
28	pg_cron/pgAgent	SQL Agent
29	pgaudit	SQL Audit
30	PostgreSQL HA/DR	AG/FCI/Log Shipping
31	Cloud PostgreSQL	RDS/Azure PostgreSQL
32	PostgreSQL upgrades	SQL Server version upgrades

Priority Focus for an Experienced SQL Server DBA

For an experienced SQL Server DBA, generic database concepts do not need to be relearned from scratch. Concentrate on PostgreSQL's architectural differences:

- PostgreSQL cluster architecture
- MVCC internals
- VACUUM, Autovacuum and bloat
- WAL architecture
- PITR and WAL archiving
- Streaming replication
- Replication slots
- Patroni, DCS and failover
- pg_stat_activity, pg_stat_statements and PostgreSQL performance diagnostics
- pg_hba.conf, roles and PostgreSQL security model

Final DBA Mental Model

Use SQL Server as the anchor for familiar concepts such as databases, schemas, tables, indexes, transactions, execution plans, backups, LSNs, replication and HA/DR. Then deliberately learn the PostgreSQL-specific internals—especially MVCC, VACUUM/autovacuum, WAL, replication slots, pg_hba.conf, extensions, FDWs, PgBouncer and Patroni. This approach minimizes duplicate learning while preserving the architectural differences that matter in production.